

Assignment 4: Take a sample dataset and demonstrate the logistic classifier with L1 and L2 regularization.

```
import numpy as np
import pandas as pd

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
data = load_breast_cancer()
X = data.data
y = data.target

# Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
#Logistic Regression with L2 (Ridge)
l2_model = LogisticRegression(penalty='l2', solver='lbfgs', max_iter=1000)

l2_model.fit(X_train, y_train)

y_pred_l2 = l2_model.predict(X_test)

print("L2 Accuracy:", accuracy_score(y_test, y_pred_l2))
```

L2 Accuracy: 0.9736842105263158

```
#Logistic Regression with L1 (Lasso)
l1_model = LogisticRegression(penalty='l1', solver='liblinear', max_iter=1000)

l1_model.fit(X_train, y_train)

y_pred_l1 = l1_model.predict(X_test)

print("L1 Accuracy:", accuracy_score(y_test, y_pred_l1))
```

L1 Accuracy: 0.9736842105263158

```
#Compare Coefficients
l1_zero = np.sum(l1_model.coef_ == 0)
l2_zero = np.sum(l2_model.coef_ == 0)

print("Number of zero coefficients (L1):", l1_zero)
print("Number of zero coefficients (L2):", l2_zero)
```

```
Number of zero coefficients (L1): 15
Number of zero coefficients (L2): 0
```